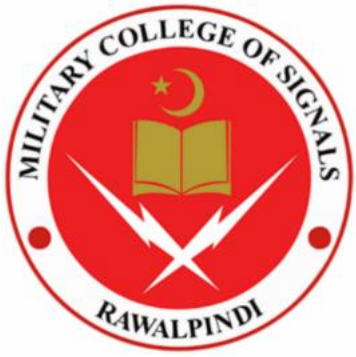




MILITARY COLLEGE OF SIGNALS

DEPARTMENT OF COMPUTER SOFTWARE ENGINEERING

Assignment - 1

Fundamentals of Computer Programming	
Submitted To	Prof. Dr. Imran Usman
Submitted By	Muhammad Rehan
Class	32 (B)

Printing stars using C++.

Task 1:

Source code:

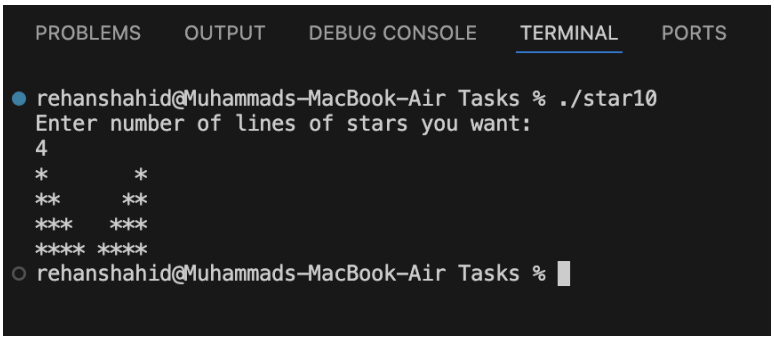
```cpp
#include <iostream>
using namespace std;

int main()
{

int n;
cout << "Enter number of lines of stars you want:\n";
cin >> n;

for (int i = 0; i < n; i++)
{
for (int j = 0; j < (i+1); j++)
{
cout << "*";
}
for(int k = i; k < (n*2)-(i+1); k++)
{
cout << " ";
}
for (int l = 0; l < (i+1); l++)
{
cout << "*";
}
cout << "\n";
}
return 0;
}
```

Output:



Taks2:

Source code:

```cpp
#include <iostream>
```

```cpp
using namespace std;

int main()
{
int n;

cout << "Enter number of lines you want star pattern:\n";
cin >> n;

for(int i = 0; i < n-1; i++)
{
for(int j = 0; j < (n-(i+1)); j++)
{
cout << " ";
}
for(int k = 0; k < (i+1); k++)
{
cout << "* ";
}
cout << "\n";
}

//Inverted pattern for completing tree
for(int i = 0; i < n; i++)
{
for(int l = 0; l < i; l++)
{
cout << " ";
}
for(int m = 0; m < (n-i); m++)
{
cout << "* ";
}
cout << "\n";
}

return 0;
}
```

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

rehanshahid@Muhammads-MacBook-Air Tasks % ./star11
Enter number of lines you want star pattern:
4
   *
  * *
 * * *
* * * *
 * * *
  * *
   *
rehanshahid@Muhammads-MacBook-Air Tasks %
```

Task3:

Source code:

```cpp
#include <iostream>
using namespace std;

int main()
{

int n;

cout << "Enter numbber of lines you want star pattern:\n";
cin >> n;

for(int i = 0; i < (n-1); i++)
{
for(int j = 0; j < (i+1); j++)
{
cout << "* ";
}
cout << "\n";
}

for(int i = 0; i < n; i++)
{
for(int j = i; j < (n); j++)
{
cout << "* ";
}
cout << "\n";
}

return 0;
}
```

Output:

